GEC HACKATHON 2026

Smart Campus – Course, Faculty & Timetable Management

Conflict-free academic scheduling for a connected campus

PROBLEM STATEMENT 04 | GEC HACKATHON 2026

1. PROBLEM BACKGROUND

Creating a college timetable requires coordinating courses, faculty availability, student groups, classrooms and time slots. Manual scheduling can create overlapping classes, double-booked rooms and uneven resource utilisation. Build a university scheduling platform that allocates academic resources while detecting and preventing timetable conflicts.

2. OBJECTIVES

- Maintain academic information for departments, courses, faculty, student groups and classrooms.
- Allocate faculty and rooms based on availability, capacity and scheduling constraints.
- Prevent overlapping assignments and make timetable information easy to access.
- Notify relevant users when an approved timetable changes.

3. FUNCTIONAL REQUIREMENTS

1.Create data models for departments, courses, faculty, student groups, classrooms and time slots.
2.Provide APIs for course registration, faculty allocation, timetable creation and timetable lookup.
3.Automate or assist faculty and classroom allocation using availability and room-capacity rules.
4.Generate events when course, faculty or classroom assignments change.
5.Validate timetable bookings and conflicts synchronously; send timetable updates and reminders asynchronously.
6.Implement separate permissions for students, faculty, coordinators and administrators.

4. EXPECTED DELIVERABLES

A working prototype with academic data management, timetable generation or allocation, conflict detection, role-based views and change notifications. Demonstrate that the system rejects or flags a faculty, classroom or student-group time clash.

5. SUGGESTED APPROACH & SCOPE

Start with a small sample campus containing a few departments, courses, faculty members, rooms and time slots. Make scheduling constraints visible and explain how the system handles cases where no valid slot is available.

6. TECHNICAL AREAS COVERED

AI / Automation • Data Modelling • API & Component Communication • Event Processing • Synchronous vs Asynchronous Workflows • Security & Access Control